

Sumit Raj

Generative AI Engineer

CONTACT

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Portfolio: sumit966-github-io.vercel.app

SKILLS

LLMs and GenAI: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, OpenAI API, Hugging Face, Transformers, LoRA
Fine-tuning, Embeddings, Agent Workflows

Retrieval: ChromaDB, FAISS, Sentence Transformers, Hybrid Search (BM25 + Dense)

Deep Learning: PyTorch, TensorFlow, Scikit-learn

Backend and Deployment: Python, SQL, FastAPI, Docker, GitHub Actions CI/CD, Git

Building: PEFT, tool calling, multi-agent systems, LLM evaluation

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur

2024 – May 2026

M.Tech, Applied AI and Machine Learning

CGPA: 6.53/10

Dr. D.Y. Patil Institute of Technology, Pune

2019 – 2023

B.E., Computer Science

CGPA: 7.89/10

CERTIFICATIONS

- DeepLearning.AI: Generative AI with LLMs (in progress)
- Coursera: Deep Learning Specialization (in progress)
- Google Cloud: Cloud Fundamentals

EXPERIENCE

Software Engineer Intern, Salesforce (Remote)

Feb 2022 – May 2022

- Built Python bulk data-processing pipelines, cutting manual data handling by approximately 40%.
- Integrated Salesforce APIs into optimized backend workflows, improving system efficiency by approximately 25%.
- Wrote SQL queries with joins and aggregations to support CRM data analysis for business teams.

PROJECTS

Multi-Agent Research Assistant

2026

Python, LangGraph, LangChain, OpenAI API, FastAPI

- Built a LangGraph multi-agent workflow in which planner, researcher and writer agents use tool calling to produce cited research reports.
- Added an evaluation step that checks each answer against its sources before returning it.
- Served the workflow through FastAPI and Docker.

Domain-Specific LLM Fine-Tuning with LoRA

2026

Python, Hugging Face, PEFT, PyTorch

- Fine-tuned an open-source LLM with LoRA on a domain question-answer dataset.
- Compared the fine-tuned model with the base model on a fixed evaluation set.
- Published training code, results and a model card on GitHub.

RAG-Based Clinical Question Answering System

2025

Python, LangChain, ChromaDB, FastAPI, Docker, GitHub Actions

- Built a RAG pipeline for clinical document retrieval with LangChain and ChromaDB, reaching 92% answer relevance on a 100-question benchmark.
- Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18%; deployed with FastAPI, Docker and GitHub Actions CI/CD at under 500 ms latency.

LLM-Powered Security Log Analyzer

2025

Python, LangGraph, OpenAI API

- Built a LangGraph agent workflow with the OpenAI API to analyze 50K+ security logs and generate natural-language threat summaries.
- Added a self-reflection evaluation loop that kept hallucination under 3% on 200 curated test cases.

Military Vehicle Detection and Face Authentication Security System

Academic Project

Python, YOLOv8, OpenCV, Haar Cascade

- Built a real-time surveillance system with Haar Cascade face authentication at approximately 95% detection accuracy.
- Detected and tracked military vehicles with YOLOv8 and OpenCV, sending automated email alerts for security events.

Brain Tumor Detection from MRI Scans

2024

Python, Deep Learning

- Trained a deep learning model to classify brain tumors on 3,000+ MRI scans.

Mobile Botnet Detection System

2023

Python, Scikit-learn

- Built a machine learning model to detect malicious bot communications with C&C servers by analyzing Android network traffic patterns.